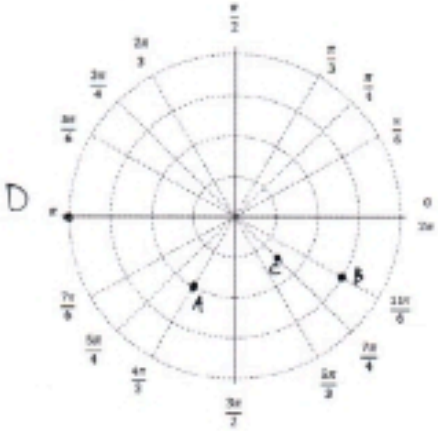
MAT061 Second Prelim Exam
FIRST SEMESTER, A.Y 2022-2023
November 15, 2022

NAME: Answer Key **Section:**_____ **Score:**_____
Signature:_____ **Instructor:**_____

GENERAL DIRECTION: Show all the necessary solutions legibly. (AVOID ERASURES)

I. Plot the following points on the given polar coordinate system. Label your points.

1. $\left(2, -\frac{2\pi}{3}\right)$ = A
2. $\left(-3, \frac{5\pi}{6}\right)$ = B
3. $\left(-\frac{3}{2}, \frac{3\pi}{4}\right)$ = C
4. $(4, -3\pi)$ = D



II. Multiple Choice. Write the letter corresponding to the right answer on the space provided for. (**2 points** each)

q 1. The graph of the curve defined by the parametric equations $x = t - t^2$ and $y = t + 1$ is a
(p) circle (q) parabola (r) line (s) ellipse

p 2. The graph of the curve defined by $r \sin\theta = -4$ is a
(p) line parallel to the polar axis (q) line parallel to the $\frac{\pi}{2}$-axis
(r) line parallel to and above the polar axis (s) line parallel and to the left of $\frac{\pi}{2}$-axis

r 3. The polar equation $4r\cos\theta + r\sin\theta = 8$ when expressed in rectangular equation is
(p) $4x + y + 8 = 0$ (q) $x + 4y - 8 = 0$ (r) $y = 8 - 4x$ (s) none of the above

q 4. The rectangular equation $x^2 = 9 - y^2$ into a polar equation is
(p) $r = 3\sin\theta$ (q) $r = 3$ (r) $r\sin\theta + r\cos\theta = 3$ (s) $r\sin\theta + r\cos\theta = 9$

q 5. What is $\left(4, -\frac{\pi}{6}\right)$ in rectangular coordinates?
(p) $(2\sqrt{3}, 2)$ (q) $(2\sqrt{3}, -2)$ (r) $(2, 2\sqrt{3})$ (s) $(-2, 2\sqrt{3})$

r 6. What is $(0, -6)$ in polar coordinate form?
(p) $(-6, \pi)$ (q) $\left(6, \frac{\pi}{2}\right)$ (r) $\left(6, \frac{3\pi}{2}\right)$ (s) $(-6, 2\pi)$

III. Eliminate parameter t in the set of parametric equations $x = 3\sin t - 2$, $y = 3 - \cos t$ and identify the graph of the resulting eqution. (6 points)

$x = 3\sin t - 2 \Rightarrow \frac{x+2}{3} = \sin t \Rightarrow \left(\frac{x+2}{3}\right)^2 = \sin^2 t \Rightarrow \frac{(x+2)^2}{9} = \sin^2 t$

$y = 3 - \cos t \Rightarrow y - 3 = -\cos t \Rightarrow (y-3)^2 = (-\cos t)^2 \Rightarrow (y-3)^2 = \cos^2 t$

Thus,

$\sin^2 t + \cos^2 t = 1 \Rightarrow \frac{(x+2)^2}{9} + (y-3)^2 = 1$ wk is an ellipse

IV. Find the set of parametric equations (in standard form) of the line passing through the point $(-3, 5)$ and parallel to the line with $2x + 3y = 5$. (6 points)

Let L be the line in the problem.

Since $L \parallel 2x+3y=5$, L has slope:

$3y = -2x+5$

$y = -\frac{2}{3}x + \frac{5}{3}$

$m_L = -\frac{2}{3} = \frac{b}{a}$

Choose $b = -2$ & $a = 3$.

Since $b < 0$, $k = \frac{-1}{\sqrt{3^2+(-2)^2}} = \frac{-1}{\sqrt{13}}$

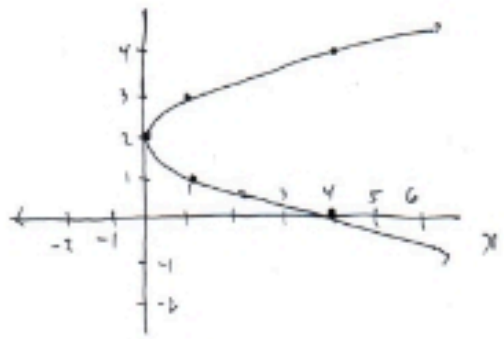
Thus, the set of parametric equations is given by:

$x = -3 + \frac{-1}{\sqrt{13}} \cdot 3t \qquad y = 5 + \frac{-1}{\sqrt{13}} \cdot -2t$

$= -3 - \frac{3}{\sqrt{13}} t \qquad = 5 + \frac{2}{\sqrt{13}} t$

V. Sketch the graph of the set of parametric equations $\begin{cases} x &= t^2 + 2t + 1 \\ y &= 1 - t. \end{cases}$ (6 points)

t	-3	-2	-1	0	1	2	3
x	4	1	0	1	4	9	16
y	4	3	2	1	0	-1	-2



VI. Find the equations of the tangent line and normal line to the curve $x = \ln(1-t)$, $y = e^{2t}$ at $t = 0$. (8 points)

$$\frac{dx}{dt} = \frac{1}{(1-t)} \cdot -1 = \frac{-1}{1-t}$$

$$\frac{dy}{dt} = e^{2t} \cdot 2 = 2e^{2t}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2e^{2t}}{\frac{-1}{1-t}} = -2e^{2t}(1-t)$$

when $t=0$,

$$m_{LT} = f'(0) = -2e^{2(0)}(1-0) = -2$$

$$m_{LN} = 1/2$$

when $t=0$

$$x = \ln(1-0) = \ln 1 = 0$$

$$y = e^{2(0)} = e^0 = 1$$

Thus, the eq'n of the tangent line is

$$y-1 = -2(x-0)$$

$$y-1 = -2x$$

$$\boxed{2x+y-1=0}$$

while the equation of the normal line is

$$y-1 = \frac{1}{2}(x-0)$$

$$2y-2 = x$$

$$\boxed{x-2y+2=0}$$

VII. Find $\frac{d^2y}{dx^2}$ of the set of parametric equations $x = e^{1-t}, y = 1 - 2t + t^2$. (6 points)

$$\frac{dx}{dt} = e^{1-t}(-1) = -e^{1-t}$$

$$\frac{dy}{dt} = -2+2t$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2+2t}{-e^{1-t}} = (2-2t)e^{t-1}$$

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = (2-2t)e^{t-1} + e^{t-1}(-2)$$

$$= (2-2t)e^{t-1} - 2e^{t-1}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{(2-2t)e^{t-1} - 2e^{t-1}}{-e^{1-t}} \quad \text{or} \quad (-2+2t)e^{2(t-1)} + 2e^{2(t-1)}$$

VIII. Find the length of each given arc.

1. $C = \{(x, y) : x = \frac{1}{6}y^3 + \frac{1}{2}y^{-1}, 1 \le y \le 2\}$ (10 points)

$f(y) = \frac{1}{6}y^3 + \frac{1}{2}y^{-1}$

$f'(y) = \frac{1}{6} \cdot 3y^2 + \frac{1}{2}(-1)y^{-2}$

$= \frac{y^2 - y^{-2}}{2}$

$[f'(y)]^2 = \left(\frac{y^2 - y^{-2}}{2}\right)^2$

$= \frac{y^4 - 2y^{2-2} + y^{-4}}{4}$

$= \frac{y^4 - 2 + y^{-4}}{4}$

$S = \int_1^2 \sqrt{1 + \frac{y^4 - 2 + y^{-4}}{4}}\, dy$

$= \int_1^2 \sqrt{\frac{4 + y^4 - 2 + y^{-4}}{4}}\, dy$

$= \frac{1}{2}\int_1^2 \sqrt{y^4 + 2 + y^{-4}}\, dy$

$= \frac{1}{2}\int_1^2 \sqrt{(y^2 + y^{-2})^2}\, dy$

$= \frac{1}{2}\int_1^2 (y^2 + y^{-2})\, dy$

$= \frac{1}{2}\left[\frac{y^3}{3} + \frac{y^{-1}}{-1}\right]_1^2$

$= \frac{1}{2}\left(\left[\frac{2^3}{3} + \frac{2^{-1}}{-1}\right] - \left[\frac{1^3}{3} - 1\right]\right)$

$= \frac{1}{2}\left(\left[\frac{8}{3} - \frac{1}{2}\right] - \left[\frac{1}{3} + 1\right]\right)$

$= \frac{1}{2}\left[\frac{8}{3} - \frac{1}{2} - \frac{1}{3} + 1\right]$

$= \frac{1}{2} \cdot \frac{16 - 3 - 2 + 6}{6}$

$= \frac{17}{12}$

2. $C = \{(x, y) : x = 2(2t+3)^{3/2}, y = 3(t+1)^2, -1 \le t \le 3\}$ (10 points)

$x'(t) = 2 \cdot \frac{3}{2}(2t+3)^{1/2} \cdot 2 = 6(2t+3)^{1/2}$

$y'(t) = 3 \cdot 2(t+1) = 6(t+1)$

$\Rightarrow [x'(t)]^2 = [6(2t+3)^{1/2}]^2 = 36(2t+3)$

$[y'(t)]^2 = [6(t+1)]^2 = 36(t+1)^2$

$s = \int_{-1}^{3} \sqrt{36(2t+3) + 36(t+1)^2}\, dt$

$= \int_{-1}^{3} \sqrt{36(2t+3 + t^2+2t+1)}\, dt$

$= \int_{-1}^{3} 6\sqrt{t^2+4t+4}\, dt$

$= 6\int_{-1}^{3} \sqrt{(t+2)^2}\, dt$

$= 6\int_{-1}^{3} (t+2)\, dt$

$= 6\left[\frac{t^2}{2} + 2t\right]_{-1}^{3}$

$= 6\left(\left[\frac{3^2}{2} + 2(3)\right] - \left[\frac{(-1)^2}{2} + 2(-1)\right]\right)$

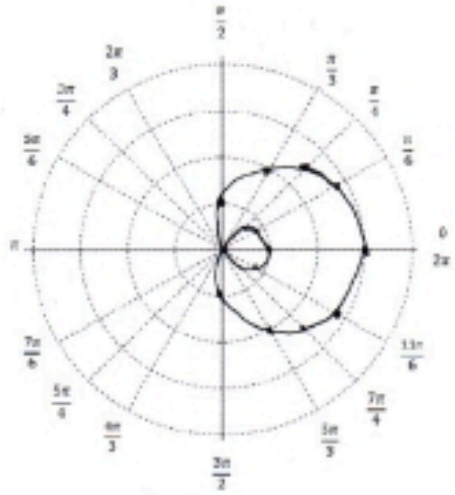
$= 6\left(\frac{9}{2} + 6 - \frac{1}{2} + 2\right)$

$= 6(4+8)$

$= 6(12)$

$= 72$

X. Discuss the symmetries and sketch the graph of $r = 1 + 2\cos\theta$. (10 points)



wrt x-axis: ($-\theta$ for θ)

$r = 1 + 2\cos(-\theta)$

$= 1 + 2\cos\theta$

$\Rightarrow$ sym wrt x-axis

wrt y-axis: ($\pi - \theta$ for θ)

$r = 1 + 2\cos(\pi - \theta)$

$= 1 + 2\left[\cos\pi\cos\theta - \sin\pi\sin\theta\right]$

$= 1 - 2\cos\theta$

$\Rightarrow$ not sym to y-axis

origin

Clearly, not symmetric to origin

θ	$\cos\theta$	$2\cos\theta$	$r = 1 + 2\cos\theta$
0 rad	1	2	3
$\pi/6$	$\sqrt{3}/2$	$\sqrt{3}$	$1+\sqrt{3} \approx 2.7$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}$	$1+\sqrt{2} \approx 2.4$
$\pi/3$	$1/2$	1	2
$\pi/2$	0	0	1
$2\pi/3$	$-1/2$	-1	0
$3\pi/4$	$-\sqrt{2}/2$	$-\sqrt{2}$	$1-\sqrt{2} \approx -0.4$
$5\pi/4$	$-\sqrt{3}/2$	$-\sqrt{3}$	$1-\sqrt{3} \approx -0.7$
π	-1	-2	-1

IX. Find the equation of the circle of curvature of the arc $y = 4x - x^2$, at $x = 1$. (8 points)

$f(x) = 4x - x^2$

$f'(x) = 4 - 2x$

$f''(x) = -2$

$$k = \frac{-2}{\left(1+(4x-2x)^2\right)^{3/2}}$$

at $x=1$,

$$k(1) = \frac{-2}{(1+4)^{3/2}}$$

$$= \frac{-2}{5^{3/2}}$$

$$\Rightarrow R = \left| \frac{1}{\frac{-2}{5\sqrt{5}}} \right|$$

$$= \frac{5\sqrt{5}}{2}$$

Note that when $x=1$,

$y = 4(1) - 1^2$

$= 3$

$(1,3)$

Center of circle of curvature:

$$x = 1 - \frac{2(1+4)}{-2}$$

$$= 1 - \frac{10}{-2}$$

$$= 1+5$$

$$= 6$$

$$y = 3 + \frac{1+4}{-2}$$

$$= 3 - \frac{5}{2}$$

$$= 1/2$$

Thus, the eq'n of the circle of curvature is

$$(x-6)^2 + \left(y - \tfrac{1}{2}\right)^2 = \frac{125}{4}$$